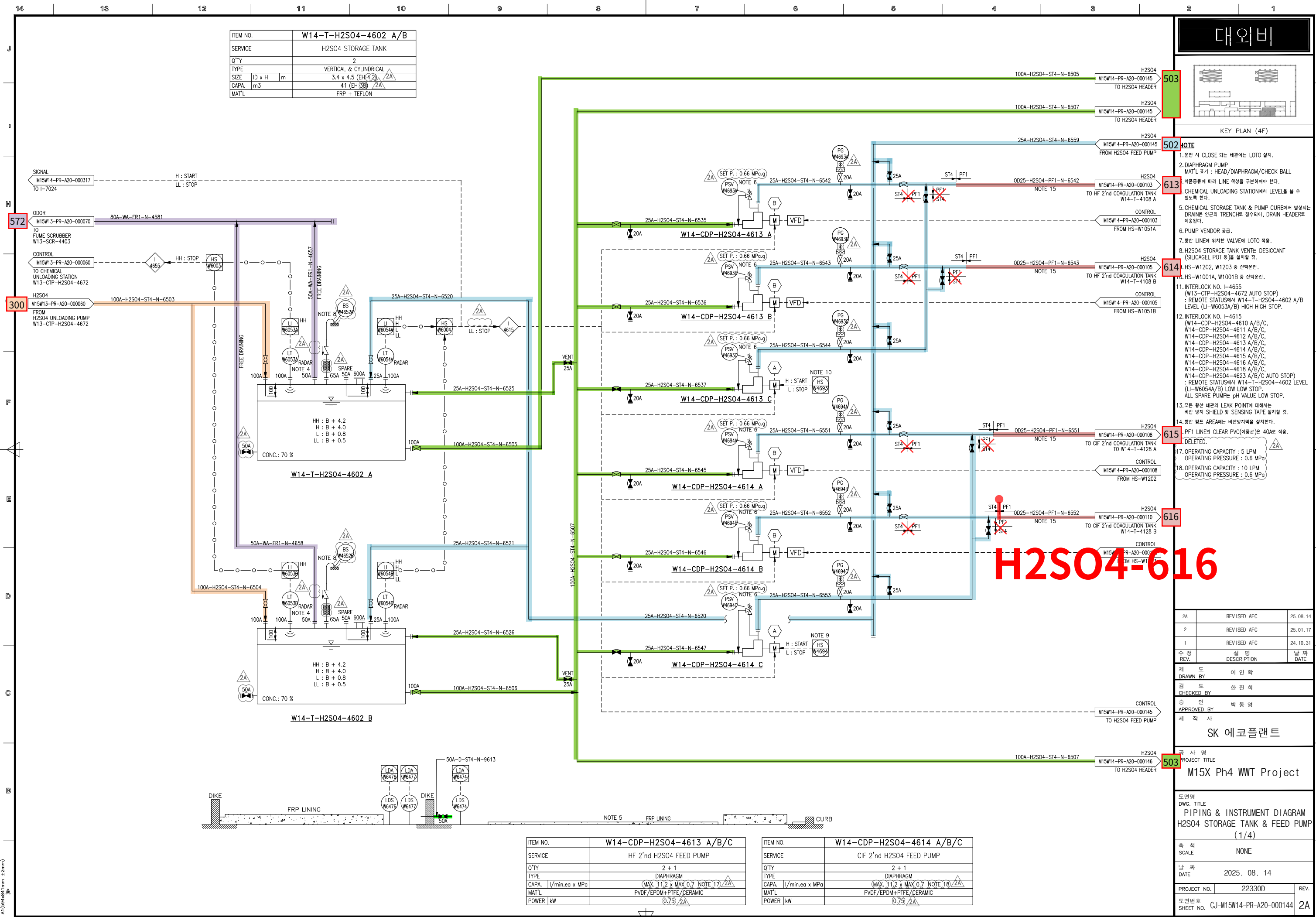


DRAWING LIST FOR TEST PACKAGE PROJECT NO.: 22328D PROJECT TITLE : M15X Ph2 Project		H2SO4-616
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PROJECT NO.: 22328D**PROJECT TITLE : M15X Ph2 Project**

H2SO4-616

[illegible]



ITEM NO.	W14-T-H2SO4-4602 A/B
SERVICE	H2SO4 STORAGE TANK
Q'TY	2
TYPE	VERTICAL & CYLINDRICAL
SIZE	3.4 x 4.5 (EH/4.2)
CAPA.	41 (EH/38)
MAT'L	FRP + TEFLON

ITEM NO.	W14-CDP-H2SO4-4613 A/B/C
SERVICE	HF 2nd H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	1 l/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 17)
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	0.75

ITEM NO.	W14-CDP-H2SO4-4614 A/B/C
SERVICE	CIF 2nd H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	1 l/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 18)
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	0.75

대외비

KEY PLAN (4F)

502

NOTE

1. 온전 시 CLOSE 되는 배관에는 LOTO 설치.
2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD/DIAPHRAGM/CHECK BALL
3. 역류종류에 따라 LINE 역상을 구분하여야 한다.
4. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
5. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAINE 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이동한다.
6. PUMP VENDOR 공급.
7. 환산 LINE에 위치한 VALVE에 LOTO 적용.
8. H2SO4 STORAGE TANK VENT는 DESICCANT (SILICAGEL POT 등)을 설치할 것.
9. HS-W1202, W1203 중 선택은전.
10. HS-W1001A, W1001B 중 선택은전.
11. INTERLOCK NO. I-4655
(W13-CTP-H2SO4-4672 AUTO STOP)
: REMOTE STATUS에서 W14-T-H2SO4-4602 A/B LEVEL (LI-W6053A/B) HIGH HIGH STOP.
12. INTERLOCK NO. I-4615
(W14-CDP-H2SO4-4610 A/B/C,
W14-CDP-H2SO4-4611 A/B/C,
W14-CDP-H2SO4-4612 A/B/C,
W14-CDP-H2SO4-4613 A/B/C,
W14-CDP-H2SO4-4614 A/B/C,
W14-CDP-H2SO4-4615 A/B/C,
W14-CDP-H2SO4-4616 A/B/C,
W14-CDP-H2SO4-4618 A/B/C,
W14-CDP-H2SO4-4623 A/B/C AUTO STOP)
: REMOTE STATUS에서 W14-T-H2SO4-4602 LEVEL (LI-W6054A/B) LOW LOW STOP.
ALL SPARE PUMPS는 pH VALUE LOW STOP.
13. 모든 환산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
14. 환산 펌프 AREA에는 비산방지막을 설치한다.
15. PF1 LINE의 CLEAR PVC(이용관)은 40A를 적용.
16. DELETED.
17. OPERATING CAPACITY : 5 LPM
OPERATING PRESSURE : 0.6 MPa
18. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.6 MPa

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정	실명	날짜
REV.	DESCRIPTION	DATE
제 도	이 인 학	
DRAWN BY		
검 토	한 진 희	
CHECKED BY		
승 인	박 동 영	
APPROVED BY		
제 작 사		
SK 에코플랜트		
공 사 명	PROJECT TITLE	
M15X Ph4 WWT Project		
도면명	DWG. TITLE	
PIPING & INSTRUMENT DIAGRAM		
H2SO4 STORAGE TANK & FEED PUMP		
(1/4)		
속 척	SCALE	
NONE		
날 짜	DATE	
2025. 08. 14		
PROJECT NO.	22330D	REV.
도면번호	SHEET NO. CJ-M15W14-PR-A20-000144	
2A		

C:\Users\WTOP\Desktop\WP&ID\251013\W[M15X Ph-4 for 인허가]\CAD FILE_0910\WCJ-M15W14-PR-A20-000144 R2 P&ID FOR H2SO4



ST4			
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW – SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW – SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD



CLIENT



1

CONTRACTOR 

PROJECT NAME

M15X Ph4 Project

COMMODITY	H2SO4
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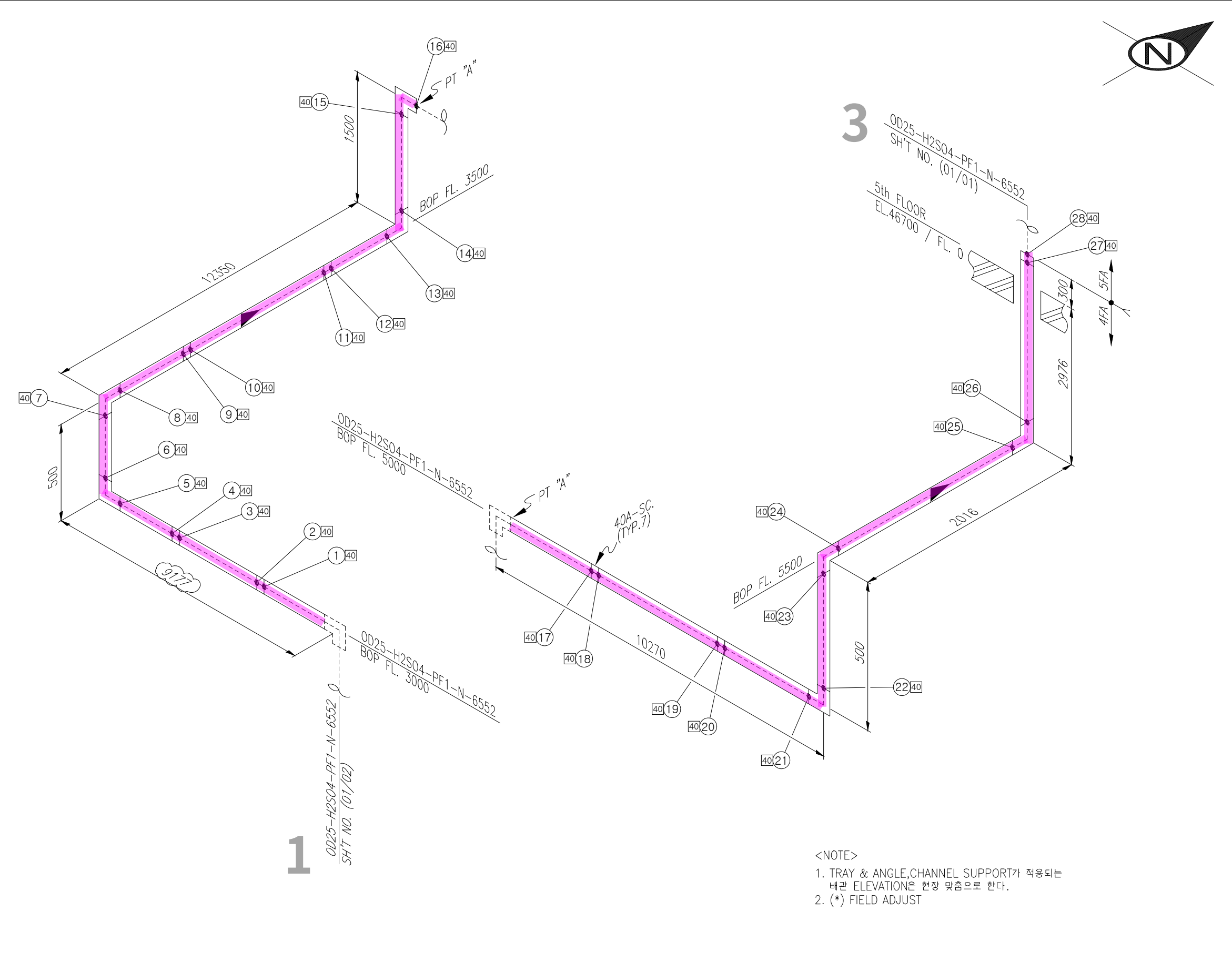
JOB NO.

22330D

LINE NO.	0D25-H2S04-PF1-N-6552
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SH'T NO. 01 / 02

V.	0
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* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME
	THICK.	—	DESIGN	6.9	60	UNIT	4FA										M15X Ph4 Project
				OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040008-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	H2SO4	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000144		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	OD25-H2SO4-PF1-N-6552	SH'T NO. 02 / 02	REV. 0

[illegible]

<NOTE>

1. (*) FIELD ADJUST
2. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.

CLIENT



3

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1O/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	6.7	60	UNIT 5FA								M15X Ph4 Project	
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050008-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY H2SO4	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000144							LINE NO.	22330D	
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	0D25-H2SO4-PF1-N-6552	SH'T NO. 01 / 01	REV. 0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



PF1			
P	OD25	45.2M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	45.2M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	11EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
SC	40A	30EA	SOCKET, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)
G1	OD25	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B1	OD25	1EA	HEX MACHINE BOLT & NYLON NUT :SM45C+HOT DIP GALV./W WASHERS
INC	25X40A	45.2M	관 보온재(고무발포) + 알루미늄 CASING

CLIENT			
		4	
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2S04		22330D	
LINE NO.	0D25-H2S04-PF1-E-6552	SH'T NO. 01 / 01	REV. 1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.9	60	UNIT RFA						
			OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동
						REF. P & ID CJ-M15W14-PR-A20-000110		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'